AIPI 590 · EMERGING TRENDS IN EXPLAINABLE AI · PROJECT 1 · POSTER COMPANION

XAI Disagreement Auditor

A per-prediction Explanation Trust Score for post-hoc XAI, and a ground-truth benchmark testing whether disagreement signals that explanations are wrong.

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THE PROBLEM

Post-hoc explainers — SHAP, LIME, Integrated Gradients — are how high-stakes model decisions get justified, and increasingly how organisations demonstrate compliance. Run several on the **same** prediction and they routinely disagree: different top features, different rankings, sometimes opposite directions. Krishna et al. (2022) named this the **disagreement problem** and found practitioners resolve it by intuition or by defaulting to a favourite tool. Nobody is told **when** an explanation is unreliable: a single SHAP plot looks equally authoritative whether every method agrees with it or none do.

WHAT I BUILT

Five attribution methods run on every prediction; six pairwise agreement metrics (Krishna et al.) score all ten method pairs; the result is an **Explanation Trust Score, 0–100**, plus a list of **contested** features — those in some methods' top-k but not others', or pushed in opposite directions. All five methods are implemented from scratch in NumPy so they share one model, one background sample and one baseline, making disagreement methodological rather than an artefact of library defaults. Unit tests confirm each is exact on linear models and that the Shapley estimator matches the reference `shap` library at $r > 0.99$.

Correctness is checked two independent ways: **faithfulness** (PGI, PGU and deletion curves against a random-attribution control — does the explanation match the **model**?) and **ground truth** (synthetic data whose real drivers are known — does it match **reality**?).

HOW TO READ THE POSTER

PANEL	WHAT IT SHOWS
1	The problem, the three research questions, the five methods audited
2 · the hook	One loan applicant, five methods. Each cell is that method's **rank** for a feature. debt_to_income is #1 for three methods and #9 for a fourth; customer_id_hash, a meaningless identifier by construction, reaches the top 2 for two methods. Trust Score 56/100.
3	The four-step pipeline, the Trust Score formula, the two verification routes, the delivered artifact
4 · main result	Trust Score vs. true-driver precision across 400 predictions
5	Agreement vs. feature correlation ρ, by model family
6	Which method actually recovers the true drivers
7	Four models at matched accuracy, 24 points apart in explanation reliability
8 – 9	Limitations, misuse risks, stakeholders, future work

FINDINGS

r = +0.61

Disagreement predicts error. Across 400 predictions, the Trust Score — computed without any ground truth — tracks how often the methods identify genuinely informative features. Low agreement is a usable warning that the explanations are wrong, which is the signal practitioners currently lack.

2 · Correlation erodes agreement. As pairwise feature correlation rises from 0 to 0.95, mean trust falls ≈ 10 points and explanation accuracy ≈ 17 points for tree ensembles and the neural network. Logistic regression is unmoved at ≈ 94. Interpretability is a property of the model, not only of the tool.

3 · The method matters. True-driver precision: LIME 84%, Shapley 81%, Integrated Gradients 79%, Occlusion 76%, SmoothGrad × Input 74%. Local gradients fail on tree ensembles, which are piecewise constant, so the gradient is near-zero almost everywhere.

Evidence base. Controlled benchmark: 8 synthetic features of which 4 are informative by construction, ρ swept over 5 levels × 4 model families × 20 predictions = 400 predictions and 2,000 explanations, all seeded and reproducible with one command.

LIMITATIONS, MISUSE RISKS AND STAKEHOLDERS

- **Agreement is not truth.** All five methods can agree and still be wrong, notably when the model genuinely relies on a proxy. The score is a warning light, not a certificate.
- **Circular evaluation.** The removal-based faithfulness tests use the same removal operation as the Occlusion method, so they favour it; this is reported inside the tool rather than claimed as a win. Scope is tabular binary classification, and the 70 / 50 bands are reasoned, not calibrated.
- **Explanation-washing** is the likeliest misuse: quoting a high Trust Score as evidence that a model is fair or compliant. It certifies only that methods **concur**. Sharper still — if a model relies on a proxy for a protected attribute, faithful methods correctly agree on it and the score goes **up**, so agreement can never be read as evidence of fairness.
- **Stakeholders.** Data scientists gain a reason to choose a method rather than a habit; auditors and compliance teams gain a per-decision flag for "do not present a single explanation here"; affected people deserve a reason for a decision that does not depend on which library the analyst imported.

ARTIFACT AND FUTURE WORK

Six-page interactive dashboard, REST API, CLI benchmark, CSV upload to audit any tabular model, 33 tests, Docker, CI, and methodology plus responsible-use documentation. Every figure on the poster regenerates from the live code with `./run.sh bench`. Next: calibrate the bands by isotonic regression so the score becomes a stated error rate; conditional (on-manifold) Shapley values to test whether the correlation effect is an artefact of off-manifold sampling; seed-to-seed stability as a sixth signal; and a human study, since the disagreement problem was identified through practitioner interviews and the remedy deserves the same scrutiny.

WHY IT MATTERS

Explainability is becoming a regulatory expectation, yet explanations ship with no reliability estimate attached — every other measurement discipline reports uncertainty; XAI mostly does not. This project contributes a per-prediction reliability signal that needs no ground truth, evidence that the signal is meaningful, and an open tool that reduces the check to a single upload.

Key references. Krishna et al. (2022), *The Disagreement Problem in Explainable Machine Learning*, arXiv:2202.01602 · Lundberg & Lee (2017), NeurIPS · Štrumbelj & Kononenko (2014), KAIS · Ribeiro et al. (2016), KDD · Sundararajan et al. (2017), ICML · Smilkov et al. (2017), arXiv:1706.03825 · Agarwal et al. (2022), *OpenXAI*, NeurIPS D&B.

Data. Seeded synthetic generators included in the repository; UCI Breast Cancer Wisconsin and Efron et al. diabetes via scikit-learn. **AI use.** Development assisted by Claude (Anthropic), per course policy.